Lecture 5: Angles and Trigonometric Functions

Trigonometry began as the study of triangles, but its functions describe anything that rotates or repeats: wheels, waves, tides, and sound. This lecture begins with right triangles and the exact values at 30°, 45°, and 60°, then defines angles of any size, extends sine, cosine, and tangent to all angles using a point on a circle, and studies their signs, symmetries, and basic identities.

1. Right-Triangle Trigonometry and Special Triangles

For an acute angle θ in a right triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

These ratios depend only on θ, since right triangles with the same acute angle are similar. Two triangles give exact values.

The 45°-45°-90° triangle. Cut a square of side 1 along a diagonal. The two legs are 1 and 1, and by the Pythagorean theorem the hypotenuse is $\sqrt{1^2+1^2} = \sqrt{2}$. The acute angles are equal, each 45°. Therefore

$$\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}, \qquad \cos 45^\circ = \frac{\sqrt{2}}{2}, \qquad \tan 45^\circ = 1.$$

The 30°-60°-90° triangle. Start with an equilateral triangle of side 2, whose angles are all 60°. Drop the altitude from one vertex to the opposite side. It bisects the base and the vertex angle, so it creates a right triangle with angles 30° and 60°, hypotenuse 2, shorter leg 1 (half the base), and longer leg $\sqrt{2^2-1^2} = \sqrt{3}$. The shorter leg is opposite 30° and the longer leg is opposite 60°, so

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3},$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \tan 60^\circ = \sqrt{3}.$$

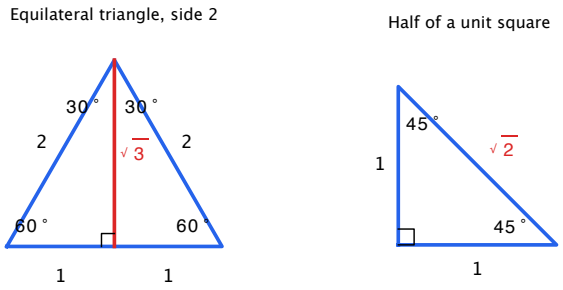


Figure 1: Special right triangles: half of an equilateral triangle of side 2 gives sides 1, root 3, 2; half of a unit square gives sides 1, 1, root 2

Triangle	Side ratios (short leg : long leg : hypotenuse)
45°-45°-90°	$1 : 1 : \sqrt{2}$
30°-60°-90°	$1 : \sqrt{3} : 2$

Notice $\sin\theta = \cos(90^\circ - \theta)$: the side opposite one acute angle is adjacent to the other.

Example. A ladder of length 12 leans against a wall at 60° with the ground. It reaches height $12\sin 60^\circ = 6\sqrt{3}$ and its base is $12\cos 60^\circ = 6$ from the wall.

Exercise 1. A right triangle has an angle of 30° and hypotenuse 10. Find the other two sides.

Exercise 2. An equilateral triangle has side s. Show that its altitude is $\dfrac{\sqrt{3}}{2}s$ and its area is $\dfrac{\sqrt{3}}{4}s^2$.

Exercise 3. Evaluate without a calculator: $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$, $\tan^2 45^\circ - \sin^2 30^\circ$, $\dfrac{\tan 60^\circ}{\tan 30^\circ}$.

Exercise 4. From a point 50 m from the base of a tower, the angle of elevation to the top is 60°. Find the height of the tower. If the angle were 30°, what would the height be?

Exercise 5. In right triangle ABC, $\angle C = 90^\circ$, $\angle B = 30^\circ$, and $AC = 1$. Extend side CB past B to a point D with $BD = AB$. Show that $\angle ADC = 15^\circ$, and use triangle ACD to find $\tan 15^\circ$ and $\sin 15^\circ$ exactly.

Exercise 6. From a point A on level ground, the angle of elevation to the top of a tower is 30°. After walking 20 m straight toward the tower, the angle of elevation is 60°. Find the height of the tower.

2. Angles and Their Measures

An **angle** is formed by rotating a ray, called the **initial side**, about its endpoint, the **vertex**, to a final position called the **terminal side**. A counterclockwise rotation gives a **positive** angle; a clockwise rotation gives a **negative** angle.

An angle is in **standard position** when its vertex is at the origin and its initial side lies along the positive x-axis. Its terminal side then lies in a quadrant or on an axis (a **quadrantal angle**, such as 0, $\pi/2$, π, $3\pi/2$).

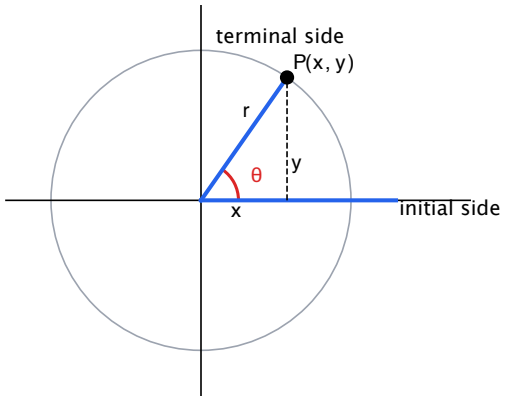


Figure 2: An angle theta in standard position with terminal point P(x,y) at distance r from the origin

Degrees and radians. One full turn is 360°. In **radian** measure, the angle θ equals the length s of the arc it cuts off on a circle of radius r, divided by r:

$$\theta = \frac{s}{r}.$$

One full turn cuts off the whole circumference $2\pi r$, so it measures 2π radians. Hence

$$\boxed{180^\circ = \pi \text{ radians}}, \qquad \text{degrees} \to \text{radians: } \times \frac{\pi}{180}, \qquad \text{radians} \to \text{degrees: } \times \frac{180}{\pi}.$$

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Coterminal angles. Two angles in standard position are **coterminal** if they have the same terminal side. They differ by whole turns:

$$\theta + 360^\circ k \quad \text{or} \quad \theta + 2\pi k, \qquad k \in \mathbb{Z}.$$

Arc length and sector area (with θ in radians):

$$s = r\theta, \qquad A = \tfrac{1}{2} r^2 \theta.$$

These formulas are valid only in radians; this is the main reason radians are the natural unit in calculus.

Example. A circle has radius 6. The arc cut off by $\theta = \dfrac{2\pi}{3}$ has length $s = 6 \cdot \dfrac{2\pi}{3} = 4\pi$, and the sector has area $A = \frac{1}{2} \cdot 36 \cdot \dfrac{2\pi}{3} = 12\pi$.

Exercise 7. Convert 150° and -45° to radians, and $\dfrac{7\pi}{6}$ and $\dfrac{5\pi}{12}$ to degrees.

Exercise 8. Find the angle in $[0, 2\pi)$ that is coterminal with $-\dfrac{5\pi}{3}$ and with $\dfrac{19\pi}{4}$.

Exercise 9. A sector of a circle of radius 10 has central angle 72°. Find its arc length and area.

Exercise 10. List three angles coterminal with 135°, including one negative angle.

Exercise 11. A wheel of radius 0.4 m turns through 5 radians. How far does a point on its rim travel? Through what angle in degrees has the wheel turned?

Exercise 12. Find the radian measure of the angle between the hour and minute hands at 4:00, and the arc length cut off on a clock face of radius 15 cm.

Exercise 13. The minute hand turns through 2π radians every hour and the hour hand through 2π radians every 12 hours. Find each hand's angular speed in radians per minute, and find the first time after 3:00 when the two hands point in the same direction.

Exercise 14. A sector of a circle has perimeter 20 (two radii plus the arc). Write its area as a function of the radius r, and find the radius and central angle that give the largest possible area.

3. Trigonometric Functions of Any Angle

Let θ be an angle in standard position and let $P(x, y) \neq (0, 0)$ be any point on its terminal side, with distance

$$r = \sqrt{x^2 + y^2} > 0$$

from the origin. The six **trigonometric functions** are

$$\sin\theta = \frac{y}{r}, \qquad \cos\theta = \frac{x}{r}, \qquad \tan\theta = \frac{y}{x},$$

$$\csc\theta = \frac{r}{y}, \qquad \sec\theta = \frac{r}{x}, \qquad \cot\theta = \frac{x}{y}.$$

A ratio is undefined when its denominator is 0. The values do not depend on which point P is chosen, because different points on the same ray give similar triangles and hence equal ratios.

Unit circle. Choosing P on the circle $x^2 + y^2 = 1$ gives $r = 1$, so

$$\boxed{\cos\theta = x, \qquad \sin\theta = y}, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}.$$

The point on the unit circle at angle θ is $(\cos\theta, \sin\theta)$.

Example. The point $P(-3,4)$ lies on the terminal side of θ. Then $r = \sqrt{9+16} = 5$, so

$$\sin\theta = \frac{4}{5}, \quad \cos\theta = -\frac{3}{5}, \quad \tan\theta = -\frac{4}{3}.$$

Special angles. From the 45°-45°-90° triangle (sides $1, 1, \sqrt{2}$) and the 30°-60°-90° triangle (sides $1, \sqrt{3}, 2$):

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

A memory aid: $\sin\theta = \frac{\sqrt{n}}{2}$ for $n = 0, 1, 2, 3, 4$ at $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, and cosine runs in reverse.

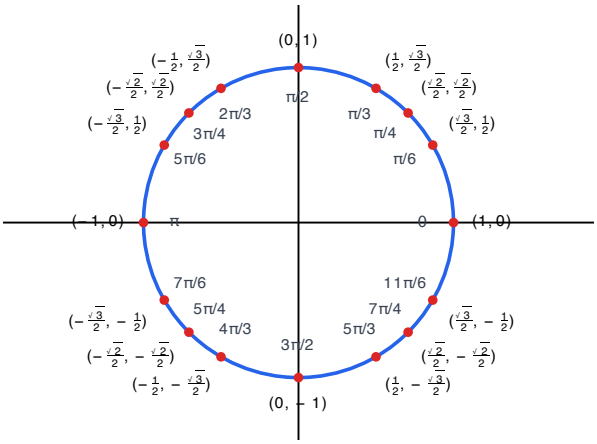


Figure 3: The unit circle with special angles in radians and their coordinates

Exercise 15. The point $(5, -12)$ lies on the terminal side of θ. Find all six trigonometric functions of θ.

Exercise 16. The point $\left(-\frac{8}{17}, y\right)$ lies on the unit circle in quadrant III. Find y and all six trigonometric functions of the corresponding angle.

Exercise 17. The terminal side of θ lies on the line $y = -2x$ in quadrant IV. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Exercise 18. Evaluate, or state that the value is undefined: $\sin\pi$, $\cos\frac{3\pi}{2}$, $\tan\frac{\pi}{2}$, $\cot\pi$, $\sec 0$.

Exercise 19. Show that the triangle with vertices $O(0,0)$, $(1,0)$, and $(\cos\theta, \sin\theta)$, where $0 < \theta < \pi$, has

area $\frac{1}{2}\sin\theta$. Use this to find the area of a regular hexagon and of a regular 12-gon inscribed in the unit circle, and compare with the area of the circle.

4. Signs of the Functions and Reduction by Symmetry

Since $r > 0$, the sign of each function is decided by the signs of x and y in the quadrant containing the terminal side:

Quadrant	x	y	sin	cos	tan
I	$+$	$+$	$+$	$+$	$+$
II	$-$	$+$	$+$	$-$	$-$
III	$-$	$-$	$-$	$-$	$+$
IV	$+$	$-$	$-$	$+$	$-$

The reciprocal functions $\csc, \sec, \cot$ share the sign of $\sin, \cos, \tan$ respectively.

ASTC rule. Going counterclockwise from quadrant I, the phrase "**A**ll **S**tudents **T**ake **C**alculus" lists which functions are positive: **A**ll in I, **S**ine (and csc) in II, **T**angent (and cot) in III, **C**osine (and sec) in IV.

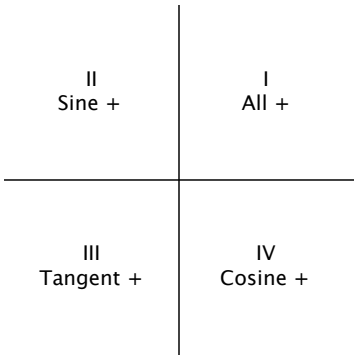


Figure 4: The ASTC diagram: all positive in quadrant I, sine in II, tangent in III, cosine in IV

Odd and even. Reflecting the terminal point (x, y) across the x-axis gives $(x, -y)$, which corresponds to the angle $-\theta$. Hence

$$\cos(-\theta) = \cos\theta \ \ \text{(even)}, \qquad \sin(-\theta) = -\sin\theta, \quad \tan(-\theta) = -\tan\theta \ \ \text{(odd)}.$$

Periodicity. Adding a full turn returns to the same terminal point:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

Tangent repeats twice as fast, since (x, y) and $(-x, -y)$ (the angles θ and $\theta + \pi$) give the same ratio:

$$\tan(\theta + \pi) = \tan\theta.$$

The **period** of sin and cos is 2π; the period of tan is π.

Reduction by symmetry. Every angle can be traced back to an acute angle in quadrant I. First use periodicity to move the angle into $[0, 2\pi)$ by adding or subtracting multiples of 2π. If the terminal side lies in quadrant I, there is nothing more to do. Otherwise let α $(0 < \alpha < \frac{\pi}{2})$ be the acute angle between the terminal side and the x-axis, and let $P(x, y)$ be the unit-circle point at angle α. The unit-circle point at θ is the image of P under one symmetry of the circle:

- **Quadrant II: $\theta = \pi - \alpha$, reflect across the y-axis.** The reflection turns the angle α measured from the positive x-axis into the angle α measured from the negative x-axis, which is $\pi - \alpha$. The point (x, y) becomes $(-x, y)$, so

$$\sin(\pi - \alpha) = \sin\alpha, \quad \cos(\pi - \alpha) = -\cos\alpha, \quad \tan(\pi - \alpha) = -\tan\alpha.$$

- **Quadrant III: $\theta = \pi + \alpha$, half-turn about the origin.** Rotating by half a turn adds π to the angle and sends (x, y) to the opposite point $(-x, -y)$ (central symmetry through the origin), so

$$\sin(\pi + \alpha) = -\sin\alpha, \quad \cos(\pi + \alpha) = -\cos\alpha, \quad \tan(\pi + \alpha) = \tan\alpha.$$

- **Quadrant IV: $\theta = 2\pi - \alpha$, reflect across the x-axis.** This is the odd/even reflection above: (x, y) becomes $(x, -y)$, the point at angle $-\alpha$, which is coterminal with $2\pi - \alpha$. So

$$\sin(2\pi - \alpha) = -\sin\alpha, \quad \cos(2\pi - \alpha) = \cos\alpha, \quad \tan(2\pi - \alpha) = -\tan\alpha.$$

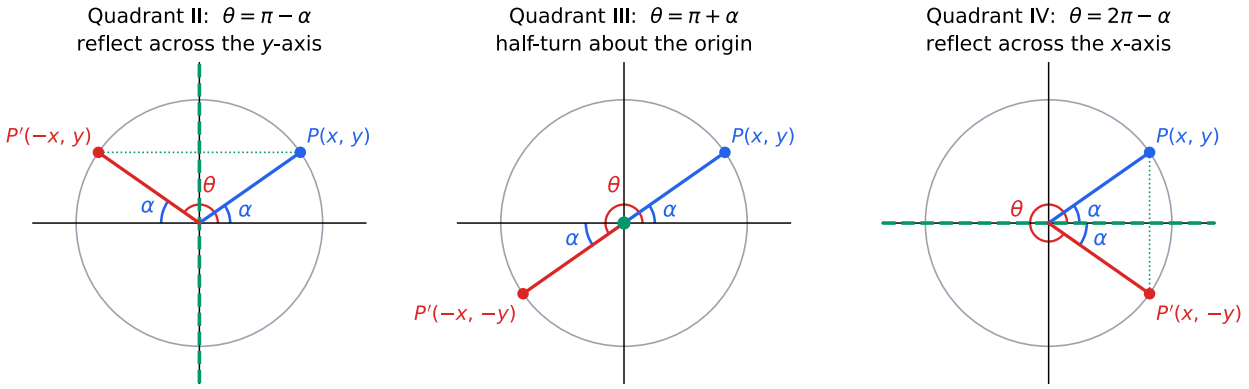


Figure 5: Reduction by symmetry: the first-quadrant point P at angle alpha is reflected across the y-axis (quadrant II, angle pi minus alpha), rotated a half-turn about the origin (quadrant III, angle pi plus alpha), or reflected across the x-axis (quadrant IV, angle 2 pi minus alpha)

Quadrant	α	Symmetry	Point	$\sin\theta$	$\cos\theta$	$\tan\theta$
II	$\pi - \theta$	across the y-axis	$(-x, y)$	$\sin\alpha$	$-\cos\alpha$	$-\tan\alpha$
III	$\theta - \pi$	half-turn about O	$(-x, -y)$	$-\sin\alpha$	$-\cos\alpha$	$\tan\alpha$
IV	$2\pi - \theta$	across the x-axis	$(x, -y)$	$-\sin\alpha$	$\cos\alpha$	$-\tan\alpha$

Each symmetry changes only the signs of the coordinates, never their sizes. So every trigonometric value at θ equals the value at α up to a sign, and the sign is the one the quadrant requires (ASTC). In practice: reduce into $[0, 2\pi)$, find the quadrant and α, evaluate at α, then attach the sign.

Example. Find sin, cos, and tan at $\dfrac{5\pi}{6}$, $\dfrac{4\pi}{3}$, and $\dfrac{7\pi}{4}$.

- $\dfrac{5\pi}{6} = \pi - \dfrac{\pi}{6}$ is in quadrant II. Reflecting $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ across the y-axis gives $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, so

$$\sin\frac{5\pi}{6} = \frac{1}{2}, \qquad \cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \qquad \tan\frac{5\pi}{6} = -\frac{\sqrt{3}}{3}.$$

- $\dfrac{4\pi}{3} = \pi + \dfrac{\pi}{3}$ is in quadrant III. The half-turn sends $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ to $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$, so

$$\sin\frac{4\pi}{3} = -\frac{\sqrt{3}}{2}, \qquad \cos\frac{4\pi}{3} = -\frac{1}{2}, \qquad \tan\frac{4\pi}{3} = \sqrt{3}.$$

- $\dfrac{7\pi}{4} = 2\pi - \dfrac{\pi}{4}$ is in quadrant IV. Reflecting $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ across the x-axis gives $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, so

$$\sin\frac{7\pi}{4} = -\frac{\sqrt{2}}{2}, \qquad \cos\frac{7\pi}{4} = \frac{\sqrt{2}}{2}, \qquad \tan\frac{7\pi}{4} = -1.$$

Example. Find $\sin\left(-\dfrac{2\pi}{3}\right)$ and $\cos 510^\circ$.

- $-\dfrac{2\pi}{3} + 2\pi = \dfrac{4\pi}{3} = \pi + \dfrac{\pi}{3}$, in quadrant III, so $\sin\left(-\dfrac{2\pi}{3}\right) = -\sin\dfrac{\pi}{3} = -\dfrac{\sqrt{3}}{2}$.
- $510^\circ - 360^\circ = 150^\circ = 180^\circ - 30^\circ$, in quadrant II, so $\cos 510^\circ = -\cos 30^\circ = -\dfrac{\sqrt{3}}{2}$.

Example. Given $\sin\theta = \dfrac{3}{5}$ and $\cos\theta < 0$, find $\cos\theta$ and $\tan\theta$. Sine positive and cosine negative place θ in quadrant II. A reference triangle has $y = 3$, $r = 5$, so $|x| = 4$, and since $x < 0$,

$$\cos\theta = -\frac{4}{5}, \qquad \tan\theta = -\frac{3}{4}.$$

Exercise 20. Determine the quadrant of θ if $\sin\theta < 0$ and $\tan\theta > 0$. If $\cos\theta = \dfrac{5}{13}$ and $\sin\theta < 0$, find $\sin\theta$ and $\tan\theta$.

Exercise 21. Evaluate exactly: $\sin\dfrac{7\pi}{4}$, $\cos\dfrac{2\pi}{3}$, $\tan\dfrac{11\pi}{6}$, $\sin\left(-\dfrac{\pi}{3}\right)$, $\cos 210^\circ$.

Exercise 22. Evaluate exactly: $\sin\dfrac{5\pi}{4}$, $\cos\dfrac{11\pi}{6}$, $\tan\dfrac{3\pi}{4}$, $\sin 300^\circ$, $\cos\dfrac{17\pi}{3}$, $\tan\left(-\dfrac{7\pi}{6}\right)$.

Exercise 23. Without a calculator, state the sign of $\sin 200^\circ$, $\cos 300^\circ$, $\tan\dfrac{5\pi}{6}$, and $\sec\left(-\dfrac{\pi}{5}\right)$.

Exercise 24. Given $\tan\theta = -\dfrac{5}{12}$ and $\sin\theta > 0$, find the other five trigonometric functions of θ.

Exercise 25. Use symmetry, not a calculator, to evaluate

$$\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \cdots + \sin 359^\circ \quad \text{and} \quad \sum_{k=0}^{11} \cos\frac{k\pi}{6}.$$

Exercise 26. Find all θ in $[0, 2\pi)$ with $2\sin^2\theta - \sin\theta - 1 = 0$.

5. Basic Properties and Identities

Pythagorean identity. Since $(\cos\theta, \sin\theta)$ lies on the unit circle $x^2 + y^2 = 1$,

$$\boxed{\sin^2\theta + \cos^2\theta = 1}.$$

Dividing by $\cos^2\theta$ or $\sin^2\theta$ gives

$$1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Reciprocal and quotient identities.

$$\csc\theta = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}.$$

Range. For every real θ,

$$-1 \le \sin\theta \le 1, \qquad -1 \le \cos\theta \le 1,$$

because x and y are coordinates on the unit circle. Tangent takes every real value.

Odd/even and periodicity were established in Section 4. Together with the cofunction relations

$$\sin\left(\frac{\pi}{2}-\theta\right)=\cos\theta, \qquad \cos\left(\frac{\pi}{2}-\theta\right)=\sin\theta,$$

they give most simplifications.

Example. If $\sin\theta=-\dfrac{2}{3}$ and θ is in quadrant III, find $\cos\theta$. From $\cos^2\theta=1-\dfrac{4}{9}=\dfrac{5}{9}$ and cosine negative in III,

$$\cos\theta=-\frac{\sqrt{5}}{3}.$$

Example (simplifying). Show $\dfrac{1-\cos^2\theta}{\sin\theta}=\sin\theta$. Since $1-\cos^2\theta=\sin^2\theta$, the left side is $\dfrac{\sin^2\theta}{\sin\theta}=\sin\theta$ for $\sin\theta\neq 0$.

Exercise 27. If $\tan\theta=2$ and $\pi<\theta<\dfrac{3\pi}{2}$, find $\sin\theta$ and $\cos\theta$.

Exercise 28. Simplify $\sin\theta\sec\theta\cot\theta$ and $(1+\sin\theta)(1-\sin\theta)$.

Exercise 29. Prove $\dfrac{\sec\theta-\cos\theta}{\sin\theta}=\tan\theta$ and $\dfrac{\cos\theta}{1-\sin\theta}=\dfrac{1+\sin\theta}{\cos\theta}$.

Exercise 30. Write $\cos\theta$ and $\tan\theta$ in terms of $\sin\theta=s$, assuming θ is in quadrant II.

Exercise 31. Use cofunction relations to show $\sin 20^\circ=\cos 70^\circ$, and find x if $\sin(3x)=\cos(x+10^\circ)$ with $0^\circ<x<30^\circ$.

Exercise 32. Use even/odd and periodicity to evaluate $\cos\left(-\dfrac{\pi}{3}\right)$, $\sin\left(\dfrac{13\pi}{6}\right)$, and $\tan\left(-\dfrac{5\pi}{4}\right)$.

Exercise 33. Suppose $\sin\theta+\cos\theta=\dfrac{1}{5}$ and $0<\theta<\pi$. Find $\sin\theta\cos\theta$, decide which quadrant θ is in, and then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Exercise 34. Reflecting a point across the line $y=x$ swaps its coordinates. Explain why this reflection sends the unit-circle point at angle α to the point at angle $\frac{\pi}{2}-\alpha$, and deduce the cofunction relations. Then use the quarter-turn $(x,y)\mapsto(-y,x)$ to show $\sin\left(\frac{\pi}{2}+\alpha\right)=\cos\alpha$ and $\cos\left(\frac{\pi}{2}+\alpha\right)=-\sin\alpha$.

Exercise 35. Find the largest and smallest values of $2\sin^2\theta+3\cos\theta$, and the values of $\cos\theta$ where they occur. (Hint: write the expression in terms of $c=\cos\theta$, with $-1\le c\le 1$.)